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Research article

In vitro modeling of the post-ingestion bioaccessibility of per- and polyfluoroalkyl substances sorbed to soil and house dust

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Abstract

Per- and polyfluoroalkyl substances (PFAS) are regularly found in soils and dusts, both of which can be consumed by children at relatively high amounts. However, there is little data available to model the bioaccessibility of PFAS in soils and dusts when consumed or to describe how the physiochemical properties of PFAS and soils/dusts might affect bioaccessibility of these chemicals. Because bioaccessibility is an important consideration in estimating absorbed dose for exposure and risk assessments, in the current study, *in vitro* assays were used to determine bioaccessibility of 14 PFAS in 33 sets of soils and dusts. Bioaccessibility assays were conducted with and without a sink, which was used to account for the removal of PFAS due to their movement across the human intestine. Multiple linear regression with backward elimination showed that a segmented model using PFAS chain length, number of branches, and percent total organic carbon explained 78.0%–88.9% of the variability in PFAS bioaccessibility. In general, PFAS had significantly greater bioaccessibility in soils relative to dusts and the addition of a sink increased bioaccessibility in the test system by as much as 10.8% for soils and 20.3% for dusts. The results from this study indicate that PFAS bioaccessibility in soils and dusts can be predicted using a limited set of physical chemical characteristics and could be used to inform risk assessment models.

Keywords: PFAS; bioaccessibility; house dust; soil

Per- and polyfluoroalkyl substances (PFAS) anthropogenic contaminants characterized by a polar functional group at the head of the structure along with having a fluorinated carbon tail (Buck *et al.*, 2011). Many PFAS have either known or suspected toxicological concerns related to liver and kidneys (Blake *et al.*, 2018), as well as several adverse developmental effects (Truong *et al.*, 2022). Two of the most studied groups of PFAS are perfluorinated carboxylic acids (PFCAs) and perfluorinated sulfonic acids (PFSAs), both of which have half-lives in humans falling between years to decades (Li *et al.*, 2018; Lindstrom *et al.*, 2011). Historically, PFAS were synthesized using electrochemical fluorination, which is a crude method that produces branched and linear PFAS (Lindstrom *et al.*, 2011). Due to their low surface tension and oleophobic properties, PFAS have been used in many applications including textiles, fabrics, paints, aqueous film-forming foams (AFFFs), and nonstick cookware (Glüge *et al.*, 2020).

PFAS are generally stable and have been detected in a variety of environmental matrices including soils (Brusseau *et al.*, 2020; Nguyen *et al.*, 2020; Schroeder *et al.*, 2021; Sörengård *et al.*, 2019; Washington *et al.*, 2019) and house dusts (de la Torre *et al.*, 2019; Eriksson and Karrman, 2015; Fraser *et al.*, 2013; Hall *et al.*, 2020; Karásková *et al.*, 2016; Knobeloch *et al.*, 2012; Strynar and Lindstrom, 2008; Su *et al.*, 2016; Tian *et al.*, 2016; Winkens *et al.*, 2017; Xu *et al.*, 2021; Zheng *et al.*, 2020). Because some children consume relatively large amounts of both soil and house dust compared to adults and other children (U.S. EPA, 2008), it is important to understand the extent to which ingested dust/soil sorbed PFAS becomes bioaccessible (percent mobilized) and the factors that influence PFAS bioaccessibility to better characterize the range of absorption factors that could be used in PFAS risk assessments that incorporate this exposure pathway. Bioaccessibility is an important variable within the exposure/dose continuum because, in part, it determines the extent to which the chemical becomes available for uptake into the circulatory system of the exposed individual.

To date, several variations of post ingestion bioaccessibility procedures (Lu *et al.*, 2021) have been used to generate data for soil/dust sorbed hydrophobic organic compounds (He *et al.*, 2019; Li *et al.*, 2022; Oomen *et al.*, 2000; Roberts *et al.*, 2019; Starr *et al.*, 2016; Yu *et al.*, 2012). Several recent studies have incorporated sorbent materials as a sink to simulate movement of the analytes across the intestine by removing them from the test system (Collins *et al.*, 2013; Fang *et al.*, 2021; Gao *et al.*, 2019; Li *et al.*, 2015a). In general, the inclusion of a sink results in increased bioaccessibility, particularly for highly hydrophobic analytes. However, as amphiphiles, PFAS may present a different

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mobilization profile and therefore it is useful to develop PFAS-specific bioaccessibility predictive models. A previous study measured bioaccessibility of perfluorooctanoic acid (PFOA) in food matrices (Li *et al.*, 2015b); however, there are no available studies of PFAS sorbed to ingested soils or dusts. Thus, this study aims to measure PFAS bioaccessibility in soils and dusts and use generated data to develop a PFAS-specific bioaccessibility predictive model.

In this study, we utilized a 3-compartment (saliva, gastric, and intestinal) human *in vitro* system to simulate the bioaccessibility of selected PFAS sorbed to ingested soils and house dusts. The assay was done in duplicate for each sample: first, a study system having no sink, and then a second study system containing a sink to trap mobilized PFAS. We then evaluated the physicochemical properties of the PFAS, soils, and dusts to determine their potential influence on PFAS bioaccessibility. Predictive statistical models were developed to determine a reduced set of variables that would be most useful for estimating PFAS bioaccessibility. As such, these predictive models would have broad applicability due to the limited set of attributes needed. Knowledge of these key variables help inform future laboratory-based bioaccessibility studies that evaluate the same or related environmental media and could also increase the utility of previously published studies where these influential variables have already been characterized. Ideally, these predictive statistical models could generate more realistic estimates of how much of an ingested chemical that is sorbed to organic matrices would be available for absorption, rather than assuming a default value of 100%, and better inform related input data commonly used in physiological-based pharmacokinetic models and for exposure and risk assessments.

Materials and methods

Chemicals

All methanol used for this study was Optima Grade and purchased from Fisher Scientific (Fair Lawn, New Jersey). Formic acid used to acidify water and methanol was 98% pure and acquired from Millipore Corporation (Billerica, Massachusetts). Ammonium hydroxide (ACS grade 28.0%–30.0%) was purchased from Sigma Aldrich (St Louis, Missouri). PFAS was purified from all matrices using ENVI-Carb graphitized carbon 250 mg SPE cartridges (Supelco-Sigma Aldrich, St Louis, Missouri) and Oasis WAX, 500 mg–60 μm (Waters Corporation, Milford, Massachusetts). Single use syringes were obtained from Becton Dickinson and Company (Franklin Lakes, New Jersey) and polypropylene syringe filters (0.45 μm, 25 mm) were purchased from Tisch Scientific (Cleves, Ohio). All water used in the clean-up procedures was 18.0 mΩ resistant and purified on site. Five Flexzorb (FM10, FM70, FM100, FM30K, and FM50K) 100% activated carbon cloths were purchased from Chemviron (Feluy, Belgium) and 1 granular activated carbon (F400) was donated from Calgon Carbon (Pittsburgh, Pennsylvania). All sorbents were evaluated for their respective efficacies as sinks. Properties of each cloth (eg, surface density, thickness) are in the Supplementary Table 1.

All labeled and unlabeled PFAS were purchased from Wellington Laboratories (Guelph, Canada). The linear PFSA used in this study were: potassium perfluoro-1-butanesulfonate (PFBS), sodium perfluoro-1-hexanesulfonate (PFHxS), sodium perfluoro-1-octanesulfonate (L-PFOS), sodium perfluoro-1-decanesulfonate (PFDS), sodium perfluoro-1-dodecanesulfonate (PFDoS), and the linear PFCAs were: perfluoro-*n*-butanoic acid (PFBA), perfluoro-*n*-hexanoic acid (PFHxA), perfluoro-*n*-octanoic acid (PFOA), perfluoro-*n*-decanoic acid (L-PFDA), and perfluoro-*n*-dodecanoic acid (PFDoA). The branched PFSAs and PFCAs used in this research were: sodium perfluoro-6-methylheptanesulfonate (Br-PFOS), sodium perfluoro-7-methyloctanesulfonate (Br-PFNS), perfluoro-4-methyloctanoic acid (Br-PFNA), and perfluoro-3,7-dimethyloctanoic acid (Br-PFDA).

Labeled PFSAs and PFCAs used for internal standards were: sodium perfluoro-1-[2,3,4-13C3]butanesulfonate (M3PFBS), sodium perfluoro-1-hexane[18O2]sulfonate (MPFHxS), sodium perfluoro-1-[1,2,3,4-13C4]octanesulfonate (MPFOS), perfluoro-*n*-[1,2,3,4,6-13C5]hexanoic acid (M5PFHxA), perfluoro-*n*-[1,2,3,4-13C4]octanoic acid (MPFOA), perfluoro-*n*-[1,2,3,4,5,6-13C6]decanoic acid (M6PFDA), and perfluoro-*n*-[1,2-13C2]dodecanoic acid (MPFDoA). Labeled PFSAs and PFCAs used as surrogate standards were: sodium perfluoro-1-[1,2,3-13C3]hexanesulfonate (M3PFHxS), sodium perfluoro-[13C8]octanesulfonate (M8PFOS), perfluoro-*n*-[2,3,4-13C3]butanoic acid (M3PFBA), perfluoro-*n*-[1,2-13C2]hexanoic acid (MPFHxA), perfluoro-*n*-[13C8]octanoic acid (M8PFOA), and perfluoro-*n*-[1,2-13C2]decanoic acid (MPFDA). Concentrations of the PFSAs were corrected to account for the mass of the counterion.

Soil and dust collection and characterization

The soil and house dust samples used for this work were collected during the first round of the American Healthy Homes Survey which was a nationally representative sampling of U.S. housing stock (Stout *et al.*, 2009). The methods used to collect, process, characterize, and store the soil and dust samples have been described previously (Sowers *et al.*, 2021; Starr *et al.*, 2016). In brief, house dusts were taken from the occupants vacuum cleaner bag while soils were collected from exterior locations near the residence. All samples were sieved and characterized (pH and percent moisture, nitrogen, organic carbon, and sand) then resieved to a final cut point (soils <250 μm; dusts <150 μm). After the second sieving, the black carbon content (soot) of the soils and dust was also measured.

Sample processing and analysis

Immediately prior to use, both WAX and Envi-Carb SPE cartridges were cleaned and conditioned (WAX: 8 ml—1% NH_4OH in methanol, 8 ml—2% formic acid in methanol (v:v); Envi-Carb: 3 ml—methanol). PFAS were extracted from soil and dust sediments with 3 × 3 ml of 1% NH_4OH in methanol (for a total of 9 ml per sample). The extracts were combined in 15 ml falcon tubes, then filtered through a 0.45-μm polypropylene filter. To ensure complete recovery of long-chain PFAS, the falcon tubes were rinsed 2 additional times using fresh methanol (1 ml each rinse). These rinsates were filtered through the 0.45-μm polypropylene, combined with the sediment extracts, and reduced to 1 ml under nitrogen in a water bath at 40°C. Finally, 1 ml 10% formic acid (in 18 mΩ resistant water) was added, the volume again reduced to 1 ml (nitrogen-water bath at 40°C) and the sample was loaded onto a WAX SPE cartridge.

After the soil/dust extract was loaded onto the WAX cartridge, the WAX was rinsed with 12 ml of 2% formic acid in methanol. The Envi-Carb cartridge was then placed underneath the WAX and the PFAS was eluted through both cartridges using 8 ml of 1% NH_4OH in methanol (v:v). The internal standards were added to the eluant, and the volume was reduced to 500 μl (N_2 evaporator/water bath at 40°C). Finally, 166 μl of 5 mM ammonium acetate and 334 μl of methanol were added and the samples were transferred to an autosampler vial and stored at 4.0°C.

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To extract PFAS from digestive fluids (Supplementary Table 2), the fluids were filtered (0.45-μm polypropylene filter) and 5 ml 10% formic acid in methanol was added to the filtrate. The filtered digestive fluid was loaded onto a precleaned and conditioned WAX cartridge which was then rinsed sequentially with 10 ml of 2% formic acid in water and 10 ml of 2% formic acid in methanol. The PFAS was eluted (10 ml of 1% NH_4OH in methanol) and collected in a 15-m falcon tube. The volume was reduced to 3 ml (nitrogen-water bath at 40°C), passed through the Envi-Carb cartridge and collected. An additional 3 ml methanol was passed through the Envi-Carb to elute and collect any PFAS remaining on the cartridge. Internal standards were added to the eluate and the volume was reduced to 500 μl (nitrogen-water bath at 40°C). Ammonium acetate (166 μl, 5 mM) and methanol (334 μl) were added, and the samples were put in autosampler vials and stored at 4.0°C.

The sinks were extracted by adding 8 ml of 1% NH_4OH in MeOH (enough to fully submerge the sink), then sonicating with heat for 20 min. Next, the sinks were vortexed for 4 min, centrifuged for 2 min, followed by filtration of the fluids through a 0.4-μM polypropylene filter. The extraction and filtration were repeated 3 times with 6 ml of 1% NH_4OH in MeOH, for a total of 26 ml. Internal standards were added, and the volume was reduced to 500 μl (nitrogen-water bath at 40°C). Ammonium acetate (166 μl, 5 mM) and methanol (334 μl) were added, and the samples were put in autosampler vials and stored at 4.0°C.

PFAS soil and their companion digestive fluid samples were analyzed using liquid chromatography (LC model 1100, Agilent Technologies, Santa Clara, California) coupled with an API 4000 tandem mass spectrometer (AB Sciex, Framingham, Massachusetts). All PFAS dust and digestive fluid samples were analyzed using an LC-QTOF system (AB Sciex). Instrument parameters can be found in Supplementary Tables 3 and 4).

Soil and dust screening

Prior to conducting the bioaccessibility assay, all soil and dust samples (500 mg aliquots) were screened to measure preexisting (native) PFAS concentrations. The samples were extracted, processed, and analyzed using the finalized method described in Sample Processing and Analysis.

Method performance

Due to the presence of native PFAS in some of the soils and dusts used to assess method performance, a combination of labeled and unlabeled PFAS were used. In all instances, labeled PFAS were spiked onto the soil/dust at the same concentration as the unlabeled PFAS. Percent recovery from soils and dust was determined by spiking 6 500 mg subsamples, each from 1 of 6 unique soil or dust samples. For each soil and dust, 3 replicates were spiked with 10 ng PFAS and the other 3 with 100 ng PFAS. Each set of 3 replicates was accompanied by a 500-ng blank sample that was spiked only with the solvent vehicle. The samples were extracted and analyzed using the procedures described in Sample Processing and Analysis.

Labeled standards were also spiked and used as needed for determination of the soil/dust method detection and quantitation limits (MDL, MQL). To estimate the soil and dust MDL/MQL, 4 replicates each of 4 soils and 4 dusts were fortified with 1.0, 3.0, or 5.0 ng PFAS and surrogates. The 16 samples (each matrix) were extracted, processed, then analyzed 4 times over a 4-day period. All data for each analyte at each concentration was pooled and linear regression of the standard deviations versus concentration was used to calculate the intercept. The MDL for each analyte was set at 3× the intercept, whereas the MQL was 10× the intercept (Taylor, 2018).

Selection and validation of chemical sink

A total of 5 carbons cloths and 1 granular activated carbon (see Materials and Methods) were initially screened for PFAS removal via single-point batch experiments. Approximately 0.600 g of each material was weighed and added to a 50-ml falcon tube. To that was added 40 ml of DI water, spiked with 100 ng of each PFAS, and shook for an hour in a water bath. For the selected sink, a 6-h time course was conducted ensure that sorption to sink occurs on a relevant timescale. Finally, for the selected sink, extraction recovery was found by spiking 100 ng of PFAS mixture directly onto the sink, equilibrating for 30 min. Due to poor recoveries of some of the PFAS from the sink, mass balances were determined to find the total percent recovery from the sediment, digestive fluid, and sink. This was performed for the bioaccessibility assays using soils, as they were relatively free of background PFAS when compared with levels found on dusts.

Bioaccessibility assay

We used a modification of method DIN 19738 (DIN 19738 2004) to measure bioaccessibility. DIN 19738 is a standard *in vitro* assay that uses 3 compartments (synthetic saliva, gastric fluids, and intestinal fluids) to represent the human digestive tract and our use of this method has been described in detail previously (Shen *et al.*, 2019; Starr *et al.*, 2016). Full details of the bioaccessibility assay procedure are in the Supplementary Material. For assays where a sink was used, a carbon fiber cloth (0.600 g) was added immediately after addition of the intestinal fluid. After 6 h, the cloth was removed and rinsed with water to remove any sediment. The rinsate was combined with the digestive fluid (DF) and sediment (sed) prior to centrifugation. Percent bioaccessibility was calculated using the following equation:

$$\text{Bioaccessibility (\%)} = \frac{[\text{PFAS}]_{\text{DF}} + [\text{PFAS}]_{\text{sink}}}{[\text{PFAS}]_{\text{DF}} + [\text{PFAS}]_{\text{sink}} + [\text{PFAS}]_{\text{sed}}} \quad (1)$$

Data analysis and quality control

Quality control of the bioaccessibility samples was maintained by (1) inclusion of a blank soil for every 6 spiked soil/dust samples and, (2) addition of a surrogate standard immediately prior to sample extraction. Internal standards were added to each sample during the final dry down to control for matrix effects and fluctuations in instrument performance.

The difference in %L-PFOS in soils and dusts were first compared using a paired *t* test after performing a Kolmogorov-Smirnov (K-S) test to confirm normality of data. Bioaccessibility data for soils and dusts were also compared using paired *t* tests: both for soils versus dusts (paired by household ID) and no sink versus with sink (paired by sample ID).

The bioaccessibility data were initially modeled using multiple linear regression (MLR) (PROC GLMSELECT; SAS Institute 2013) and employing the least absolute shrinkage and selection operator (LASSO) algorithm for stepwise model selection. Fundamentally, the GLMSELECT procedure is comparable with the commonly used regression (PROC REG) or general linear model (GLM) procedures. However, use of the LASSO algorithm avoids issues associated with using stepwise MLR selection techniques (eg, forward or backward) such as biased model fit statistics, undeservedly low *p* values, and exacerbation of collinearity, where present across the suite of independent variables.

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Preliminary MLR modeling was done to identify a set of statistically influential variables most useful in predicting bioaccessibility (retained variables, $p \leq .05$) for each dust and soil sample and for whether a sink was included in the test vessel (ie, 4 separate models). The study chemical variables included in the initial models were the number of carbons in the PFAS, the number of fluorinated carbons in the PFAS, the number of branches, and type of functional group (eg, carboxylate vs sulfonate) within the molecule. Soil characteristics included the percent total organic carbon (%TOC), percent clay, percent silt, percent sand, percent moisture, percent nitrogen, and pH. The %TOC was initially split into either black carbon or diffused carbon, as black carbon strongly decreases the bioaccessibility of other organic pollutants in soil (Agarwal and Bucheli, 2011; Semple *et al.*, 2013). However, negligible differences (<0.1% contribution to overall model fit statistics) were found between models when including %TOC versus using black carbon and diffused carbon as independent variables and, as a result, %TOC was used for simplicity.

Two new variables were also introduced to represent a segmented regression model because the bioaccessibility observed for the 4 and 6 carbon/fluorocarbon PFAS molecules was effectively at the same level (>90%), indicating PFAS molecules having these few carbons/fluorocarbons have limited to no influence on reducing bioaccessibility. Use of this approach allowed for a no-slope effect for the 4 and 6 carbon PFAS while also capturing any linear changes for the longer chain molecules and including all the other chemical and soil characteristics in the regression equation.

Finally, predictive models were then developed by considering a consistent, practical set of variables across all 4 models and determined by the variable's statistical significance and its presence contributed to greater than 1% of the overall model fit (as indicated by the variable partial R^2 values). And finally, a mixed-effects model (PROC MIXED) was employed to confirm whether the assumption of no correlation between bioaccessibility and sample ID was reasonable.

Results

Soil and dust collection and characterization

The soil and dust characteristics have been presented previously (Shen *et al.*, 2019; Starr *et al.*, 2016) and the mean and standard deviations are provided in Supplementary Table 5. Compared with the soils, the house dusts were more alkaline, higher in organic carbon, with a size distribution skewed toward the silt and clay fractions. Direct comparison of soils/dust pairs (matched by residence) showed that their compositions were unrelated.

Method performance

All method performance data, including method recovery, limits of detection, and limits of quantitation for dust, soil, and digestive fluid are in Supplementary Table 6. Method recoveries for all 3 matrices fell within 100% ± 20% for all PFAS at 10 and 100 ng spikes.

Selection and validation of chemical sink

Only 3 out of the 6 materials tested (FM10, FM70, FM100, FM30K, FM50K, and F400) demonstrated removal of PFAS (Supplementary Table 8). Of those 3, the only 2 materials with PFAS removal greater than 50% were FM30K (75.2% ± 12.1%) and F400 (77.3% ± 3.7%). Because FM30K and F400 performed similarly when removing PFAS from water, the final material selected was based on practical reasons. When added to the bioaccessibility assay, a granular carbon (F400) would need to be incorporated via a semiporous holder that would allow digestive fluid to pass through but would provide a barrier for the soil or dust. In contrast, a carbon cloth (FM30K) could be added directly to the digestive fluid and soil/dust and then separated easily post-assay. Thus, FM30K was selected to be the sink for PFAS. A 6-h time course using FM30K resulted in 78.9% ± 4.7% removal of all PFAS by 30 min and >90% removal of all PFAS from second hour and onward (Supplementary Table 9 and Figure 1). Mean extraction efficiency of PFAS from FM30K is provided in Supplementary Table 7. The PFAS that fell within 100% ± 30% recovered from FM30K were PFBA, PFHxA, PFOA, and L-PFDA. The remaining PFAS that demonstrated poor recoveries are discussed in more detail later.

Soil and dust screening

Soil and dust screening results are presented in Table 1. Although each soil and dust sample was fortified with 100 ng of each PFAS, any native PFAS already present in the sample contributed to the total mass and bioaccessibilities were calculated using both native and spiked PFAS. As shown in Table 1, native PFAS in dusts were present in greater variety and in higher concentrations relative to soils (Table 1). For example, 25% of the screened samples had native PFOA and PFOS masses that were more than twice the 100 ng that was spiked. Native soil PFAS consisted primarily of PFOA, Br-PFOS, and L-PFOS with maximum concentrations of 4.8, 5.4, and 21.7 ng/g, respectively, whereas maximum concentrations of PFAS in dusts (2628, PFOA; 975, Br-PFOS; 2076, PFOS ng/g) were higher and more commonly detected.

Of all the soils and dusts evaluated, there were 19 matched pairs of soils/dusts in which the soil and dust contained both Br-PFOS and L-PFOS. A paired *t* test indicated soils had significantly greater percent L-PFOS (82.4% L-PFOS in soils vs 74.3% L-PFOS in dusts; $p < .05$). Of those 19 pairs, the percent L-PFOS in soil correlated with percent black carbon ($p < .05$), whereas the percent L-PFOS in dust was not correlated with any of the measured characteristics.

PFAS bioaccessibility

Bioaccessibility of individual PFAS were calculated for all soils and dusts with and without a sink (Figure 1). In Figure 1, PFAS are listed from left to right in the same order as their chromatographic retention times from a C18 column, as a proxy for hydrophobicity (Rodowa *et al.*, 2020). In general, PFAS bioaccessibility decreases as hydrophobicity increases. For PFAS with ≤6 carbons, bioaccessibility is consistently >90%, regardless of the type of matrix (eg, soil vs dust) or the presence of a sink. It is well established that as the chain length of PFAS increases, so does sorption to sediments (Higgins and Luthy, 2006). Thus, PFAS with shorter chain lengths result in a greater amount desorbed and therefore, bioaccessible.

Given the same number of carbons, PFCAs were respectively more bioaccessible than PFSAs in soils and dusts. For example, for the dusts without a sink, PFOA was 91% bioaccessible, whereas PFOS was 61% bioaccessible. There were 4 branched PFAS included in this study: Br-PFOS (NaP6MHpS), Br-PFNA (P4MOA), Br-PFDA (P-3,7-DMOA), and Br-PFNS (ip-PFNS). Of those 4, PFOS and PFDA linear isomers were also included and could be directly compared. A *t* test indicated no significant differences between the bioaccessibility of Br-PFOS and L-PFOS ($p > .05$; Supplementary Table 10). In contrast, Br-PFDA was significantly more bioaccessible ($p < .0001$) than L-PFDA in dusts and soils with or without a sink. The branched isomer for PFDA included has an

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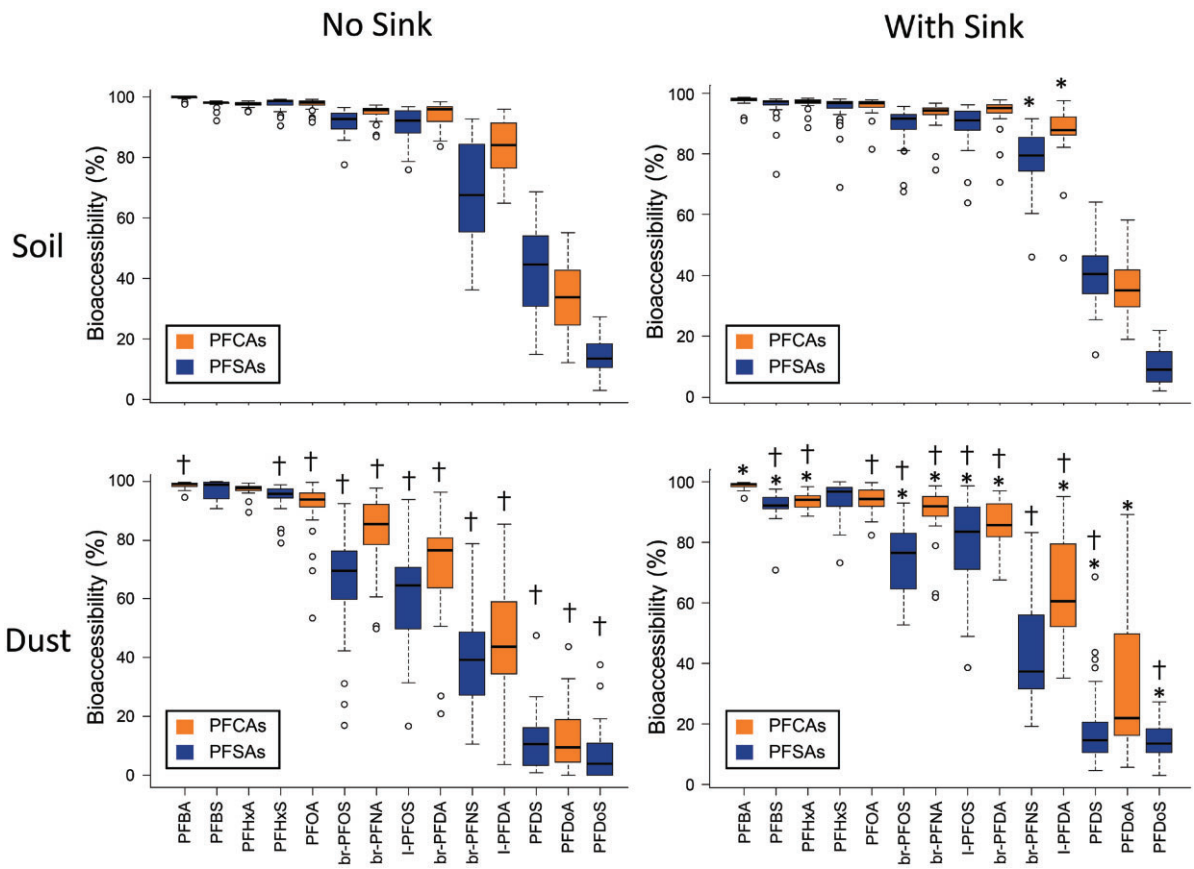

Figure 1. Bioaccessibility of PFAS in soil and dust with and without a sink. *Significantly greater bioaccessibility than no sink ($p < .05$). †Significantly lower bioaccessibility in dust relative to soil.

additional branch compared with Br-PFOS (Supplementary Figure 2), resulting in greater polarity and less hydrophobicity (Schulz *et al.*, 2020).

Generally, the mean bioaccessibilities of PFAS in soil were greater compared with dusts, with statistically significant differences ($p < .05$) for all PFAS except for most PFAS both with and without a sink. The presence of a sink significantly increased ($p < .05$) the bioaccessibility of Br-PFNS and L-PFDA in soil. The sink had a much greater impact on PFAS bioaccessibility from dusts, with significant increases for 11 out of 14 PFAS. It is possible that the sink would have had a greater impact if the initial bioaccessibilities had not already been so high, although that was not observed for the more hydrophobic PFAS in soil.

It is important to note that several of the investigated PFAS were not fully recoverable from the sink during the method performance evaluation of FM30K (Supplementary Table 7). Specifically, PFDoA, PFHxS, Br-PFOS, L-PFOS, PFDS, and PFDoS all had <50% recovery. The mass balance for individual PFAS in the bioaccessibility assays with a sink (digestive fluid, sediment, and sink) resulted in 70%–130% recovery for 10 out of 12 PFAS (Supplementary Table 11). The two exceptions were L-PFOS and Br-PFOS, which had percent recoveries 60.4% ± 5.8% and 68.8% ± 6.3%, respectively. Bioaccessibilities for these 2 PFAS were also adjusted using the percent recovery from the carbon cloth as a correction factor (Supplementary Table 7). For example, the percent recovery of Br-PFOS from the carbon cloth would be 47.2% ± 0.5% so the theoretical recovery from the bioaccessibility assay is 41.5% rather than 19.6%. Using the adjusted sink recoveries resulted in bioaccessibilities increasing from 89.8% to 93.9% (L-PFOS) and 89.8% to 92.4% (Br-PFOS). The bioaccessibilities with or without adjusting for sink recovery fell within 5% agreement with each other; thus, it is assumed that the unrecovered mass from the sink has negligible effects on the overall bioaccessibility model.

Bioaccessibility model

Four statistical models were constructed to understand the dynamic between PFAS and soil/dust characteristics and bioaccessibility: soil without a sink, soil with a sink, dust without a sink, and dust with a sink (Table 2 and Supplementary Table 12). Models for soils and dusts were separated because of their differing physical properties (Supplementary Table 5).

A multilinear model was considered appropriate in explaining variation in bioaccessibility, and mixed modeling results confirmed the assumption of collinearity between bioaccessibility and sample ID was reasonable (r values ranged from 0.07 to 0.18). Recall that due to the negligible influence to bioaccessibility of PFAS having ≤6 carbons, a modification of the input data was needed. This segmented model was created by adjusting the number of carbons in each PFAS ($^{\#}C_{adj}$). Any PFAS with ≤6 carbons was adjusted to 6 carbons, and any PFAS with >6 carbons remained the same. For example, PFBA (4 carbons) and PFDA (10 carbons) had $^{\#}C_{adj}$ values set at 6 and 10, respectively. A similar approach was taken to adjust the number of fluorinated carbons ($^{\#}CF_{n,\ adj}$) prior to generating the predictive model. In this case, if a PFAS had ≤6 fluorinated carbons, it was adjusted to 6. For example, PFHxA and PFHxS have 5 and 6 fluorinated carbons, respectively, so the $^{\#}CF_{n,\ adj}$ for both PFHxA and PFHxS was set at

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Table 1. Distribution of PFAS in soil and dust.

	PFAS	Percent detected	Concentration percentiles (ng/g) 25th	50th	75th	90th	95th	Maximum
Soil ($n = 33$)	PFBA	0	—	—	—	—	—	—
	PFHxA	0	—	—	—	—	—	—
	PFOA	85	1.0	1.4	1.8	2.6	4.1	4.8
	br-PFNA	3	—	—	—	—	—	1.6
	L-PFDA	18	—	—	—	0.6	1.0	6.7
	br-PFDA	0	—	—	—	—	—	—
	PFDoA	42	—	—	0.2	0.5	0.7	4.8
	PFBS	12	—	—	—	—	0.3	9.6
	PFHxS	3	—	—	—	—	—	2.4
	Br-PFOS	33	—	—	0.8	1.3	2.0	5.4
	L-PFOS	64	—	2.7	5.0	6.4	10.2	21.7
	br-PFNS	3	—	—	—	—	—	0.4
	PFDS	21	—	—	—	0.2	0.4	1.4
	PFDoS	0	—	—	—	—	—	—
Dust ($n = 29$)	PFBA	71	—	3.1	10.5	16	111	225
	PFHxA	90	3.4	6.0	14.0	63	254	392
	PFOA	94	5.2	10	234	1144	1222	2628
	br-PFNA	48	—	0.8	1.1	1.8	5.1	10
	L-PFDA	58	—	1.1	2.4	3.5	7.5	10
	br-PFDA	10	—	—	—	—	1.1	3.3
	PFDoA	94	1.3	1.9	4.7	5.2	5.6	10
	PFBS	94	0.3	0.6	0.8	2.0	2.9	6.8
	PFHxS	90	2.1	7.2	9.8	15	40.6	78.1
	Br-PFOS	94	4.0	6.3	12	66	592	975
	L-PFOS	94	9.0	12	172	1156	1633	2076
	br-PFNS	0	—	—	—	—	—	3.0
	PFDS	39	—	—	0.9	2.0	5.1	8.0
	PFDoS	3	—	—	—	—	—	1.3

Dashes indicate that measurements below the method detection limit (<MDL).

Table 2. Partial and total r^2 of final variables used in the 4 bioaccessibility models.

	Soil		Dust	
	No sink	With sink	No sink	With sink
$^{\#}CF_{n,\ adj}$[a]	0.664	0.623	0.768	0.621
$^{\#}br$[b]	0.135	0.202	0.072	0.045
$^{\#}C_{adj}$[c]	0.033	0.040	0.019	0.060
%TOC	0.051	0.018	0.013	0.046
Total r^2	0.883	0.882	0.872	0.772

Full model parameters can be found in Supplementary Table 12.
[a] Adjusted number of fluorinated carbons.
[b] Number of branches.
[c] Adjusted number of carbons.

6. This segmented model accounted for 77%–88% of the bioaccessibility variability. The $^{\#}C_{adj}$ and $^{\#}CF_{n,\ adj}$ together accounted for over 66% of the variability in all 4 models, emphasizing the importance of PFAS chain length and number fluorinated carbons in driving bioaccessibility. The number of branches ($^{\#}br$) was also included in all 4 models, although it had a greater impact for soils (14%–20%) as compared with dusts (4.5%–7.2%). Percent TOC was determined as the most influential soil- and dust-specific attribute and explained between 1% and 5% of variability in bioaccessibility.

Discussion

Soil and dust screening

The detection frequency of PFAS in dusts for the current study is higher than a previous study, particularly for short-chain PFAS (Hall *et al.*, 2020). For example, PFBS was detected in 94% of dusts in the current study ($n = 33$) compared with the 1.1% of dusts by Hall *et al.*, ($n = 184$). This may be due to the difference in detection limits between the 2 studies (eg, 0.14 vs 22.3 ng/g for PFBS). For all observed PFAS in dusts, the concentrations agreed with previous literature, with maximum concentrations falling around or below previous recorded maximums (Hall *et al.*, 2020). The differences in chemical profiles are represented in Figure 2, which demonstrates relative abundances of PFAS for each soil and dust. Although the scales between soils (Figure 2A) and dusts (Figure 2B) are different, there is much greater variety in PFAS composition in dusts compared with soils. This is likely due to dusts containing PFAS from additional sources such as textiles and paint (Glüge *et al.*, 2020). Alternatively, the higher abundance of short-chain PFAS in dusts relative to soils could potentially be indicative of other interactions (eg, electrostatic) playing a stronger role in retention, in addition to hydrophobic interactions.

For 17 out of 19 soils, the percent L-PFOS was >70%, which is consistent with previous work that observed linear PFAS isomers sorb to soil more readily than branched isomers (Benskin *et al.*, 2012; Schulz *et al.*, 2020). However, most dusts (13 out of 19) had enrichment of Br-PFOS, with percent L-PFOS as low as 57.7%. The difference in Br-PFOS in soils and dusts indicates that dust retains PFAS through different sorption chemistries than soils (eg, electrostatics, protein binding) that may impact the rerelease of PFAS from dust in digestive fluid.

PFAS bioaccessibility

There are no other *in vitro* bioaccessibility studies that evaluated PFOA and PFOS for comparison, however, there are multiple *in vivo* bioavailability studies. Two studies showed that PFOS was

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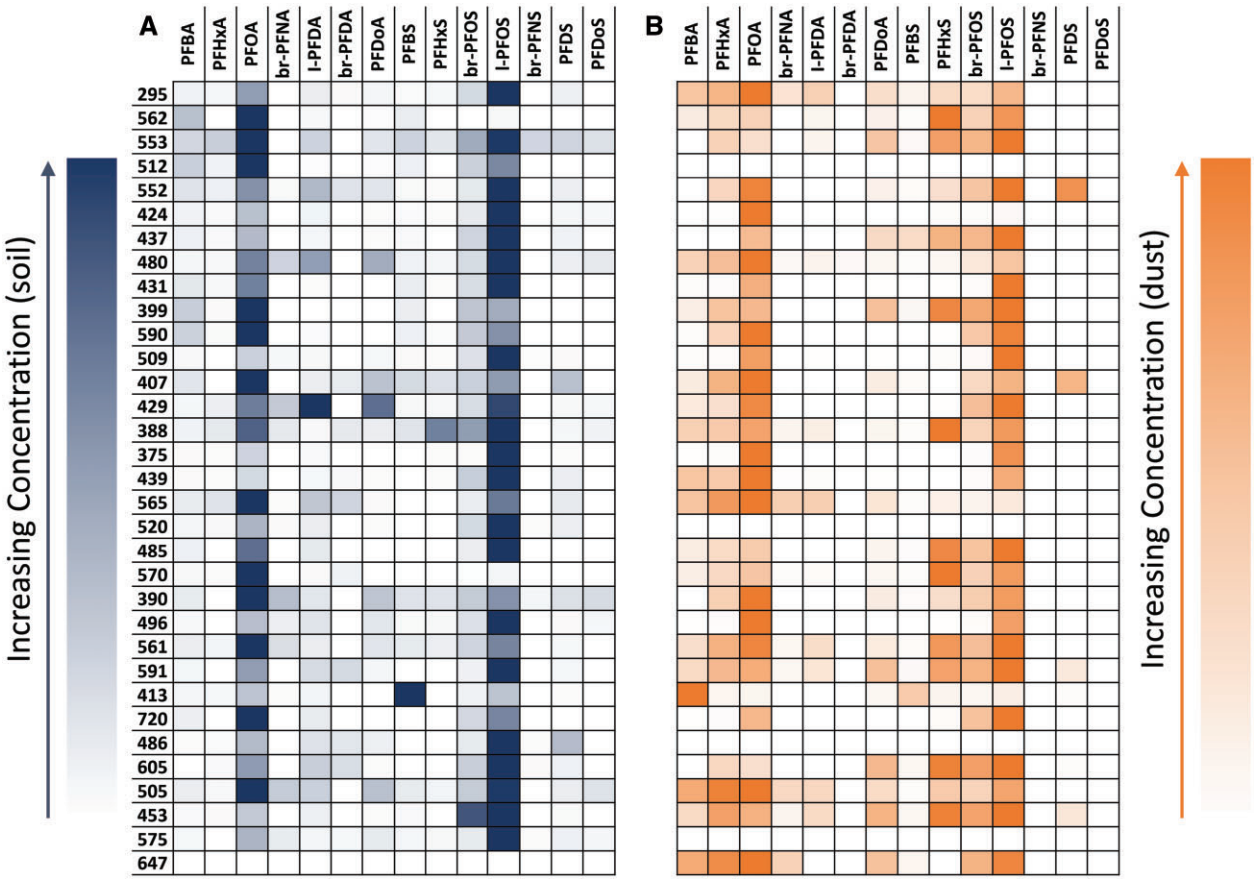

Figure 2. Relative PFAS distribution for paired soils (A) and dusts (B). Concentrations are on relative scales for each soil and dust based on the normalized concentrations of individual PFAS to the total. Numbers on the y-axis represent unique sample IDs.

more bioavailable than PFOA in earthworms exposed to biosolid-amended soil and AFFF impacted soil (Bräunig *et al.*, 2019; Wen *et al.*, 2015). The difference in *in vitro* bioaccessibility among head groups in the present study and *in vivo* relative bioavailabilities observed in previous studies could be due to differences in PFAS concentrations in soils and dusts. For instance, the concentration of PFOS in soils used by Bräunig *et al.* (2019) was as high as 13 400 ng/g, whereas the maximum concentration of PFOA was only 55.4 ng/g. For comparison, the present study had maximum concentrations of PFOS and PFOA at 2080 and 2730 ng/g, respectively (Table 1). Additionally, the greater bioaccessibility of PFOA relative to PFOS in the present study could be due to lack of a sufficient sink in the assay used in previous studies. When the assay was performed with a sink, the difference between PFOA and PFOS was lower, at 94% and 79% bioaccessible, respectively. In a previous study, the use of a sink resulted in greater agreement between *in vitro* bioaccessibility and *in vivo* bioavailability of flame retardants, indicating that a sink is more physiologically relevant (Fang and Stapleton, 2014). Future work should expand on this to compare *in vitro* bioaccessibilities of PFAS with and without a sink against *in vivo* bioavailabilities.

A difference in PFAS bioaccessibility among soils and dusts is not surprising given that there are significant differences in almost all parameters that were characterized (Supplementary Table 5). For example, dusts had significantly greater TOC than soils (18.3% ± 6.6% and 3.2% ± 2.6%, respectively). Greater TOC in sediments generally results in higher sorption of PFAS (Higgins and Luthy, 2006), which is a likely explanation for the differences in PFAS bioaccessibility observed between soils and dusts.

Bioaccessibility model

The variability accounted for using $^{\#}CF_{n,\ adj}$ is attributed to differences in head groups between PFSAs and PFCAs. For a given carbon chain length, PFSAs have an additional fluorinated carbon compared with PFCAs. There are significant differences in the physiochemical properties between fluorinated carbons and hydrocarbons (eg, electronegativity, bond length, stability) so it is unsurprising that $^{\#}CF_{n,\ adj}$ is important in the final models in addition to $^{\#}C_{adj}$ (Kiplinger *et al.*, 1994).

Dusts had significantly lower %L-PFOS than soils, with some dusts enriched in Br-PFOS (>30% Br-PFOS). It is possible that dusts have more complex retention mechanisms (eg, electrostatics) in addition to hydrophobic interactions, and that those interactions are less dependent on whether there are branches in a PFAS. However, this remains speculative, as characterizing surface chemistries for soils and dusts was outside the scope of this study.

Other variables that were statistically significant in at least one of the early model iterations are pH, %clay, %moisture, %silt, and %nitrogen. However, individually these all accounted for explaining <1% of the total variability and were removed from final models due to lack of practical significance. In fact, the only variable related to soil/dust characteristics that was included in final models was %TOC, which accounted for ≤5% of the total variability (Table 2), indicating that it is primarily the PFAS structure that determines bioaccessibility for either soils or dusts. It is possible that 12%–23% variability unaccounted for could be explained by other soil or dust properties (eg, anion exchange capacity), but that needs to be further explored.

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Chain length, branching, and %TOC were all significant factors in PFAS bioaccessibility in soils and dusts. Bioaccessibility increased with decreasing chain length, increasing branching, and decreasing %TOC. Most of the variability in bioaccessibility was due to PFAS characteristics, rather than soil or dust characteristics; indicating that bioaccessibility is largely independent of the exact soil and dust ingested. The use of a sink resulted in significant increased bioaccessibility of 3 out of 14 PFAS in soils and 11 out of 14 PFAS in dusts; thus, its possibility that assays without a sink could underestimate PFAS bioaccessibility. Future work should focus on determining the relationship between *in vitro* bioaccessibility data models and *in vivo* bioavailabilities. Using a segmented model, 77%–88% of the variability in PFAS bioaccessibility was accounted for. The results from this study indicate that PFAS bioaccessibility in soils and dusts can be predicted and could be used to inform risk assessment models in the future.

Supplementary data

Supplementary data are available at *Toxicological Sciences* online.

Declaration of conflicting interests

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